

The Spectral Weight $\pi\lambda/\sinh \pi\lambda$: Derivation from Phase Space, Modular Form, and Exact Moments

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Abstract

The function $P(\lambda) = \pi\lambda/\sinh(\pi\lambda)$ has appeared throughout this series as a spectral weight of unclear provenance. Its origin is now settled. In a companion paper the two-particle massless unitarity cut is computed in celestial variables, and its Mellin image is $\Gamma(\Delta_5)\Gamma(\Delta_6)/\Gamma(\Delta_5+\Delta_6)$ times elementary factors; restricted to the shadow locus $\Delta_5 + \Delta_6 = 2$ with $\Delta_5 = 1 + i\lambda$, this equals $\Gamma(1 + i\lambda)\Gamma(1 - i\lambda) = P(\lambda)$. So P is a phase-space factor of the cut. It is a derived quantity, and we prove here the identities it satisfies: a Planck form $P = 2\pi\lambda[n_B(\pi\lambda) - n_B(2\pi\lambda)]$ with the zero-point energies cancelling, an elementary Fourier pair with $\frac{1}{4}\text{sech}^2(x/2)$, and a ladder-moment sequence whose first two members are $\mathcal{M}_1 = 1/8$ and $\mathcal{M}_2 = 1/90 = \zeta(4)/\pi^4$, verified independently here to 26 digits.

This paper also retracts, in full and by name, the physical claims built on this function in the draft it replaces. The parallel retractions for the blackbody draft are in [6]. P is not the Plancherel density of $\text{SL}(2, \mathbb{C})$, which grows like λ^2 for the scalar principal series. P does not replace the Feynman loop measure. The bound $\mathcal{M}_L \leq (1/8)^L$ is a true statement about the convolution integrals $\mathcal{M}_L$ and a false statement about loop amplitudes, so the claim of ultraviolet finiteness without counterterms does not stand: the companion derivation reconstructs the standard loop integral exactly, divergences included. The fifteen-digit box verification reported in the earlier draft compared a Feynman-parameter integral with a reparameterization of itself. What survives is a family of exact special-function identities, and a physical interpretation of P as the spectral density of a two-particle cut rather than as a cure for anything.

Keywords: celestial holography, unitarity, Plancherel measure, modular theory, Bisognano–Wichmann, Bernoulli numbers.

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1 Where the weight comes from

Earlier drafts in this series postulated $P(\lambda)$ as a loop measure and then looked for reasons it should be one. The order of reasoning was backwards, which is how the errors listed in Sec. 4 entered. The correct order runs as follows.

Theorem 1.1 (Phase-space origin). *Let $d\Pi_2$ be two-particle massless phase space at total mass M , written in celestial coordinates. Then*

$$\int d\Pi_2 \omega_5^{\Delta_5-1} \omega_6^{\Delta_6-1} = \frac{1}{8\pi} \left(\frac{M}{2} \right)^{\Delta_5+\Delta_6-2} \frac{\Gamma(\Delta_5)\Gamma(\Delta_6)}{\Gamma(\Delta_5+\Delta_6)}, \quad (1)$$

and on the shadow locus $\Delta_5 + \Delta_6 = 2$ with $\Delta_5 = 1 + i\lambda$,

$$\frac{\Gamma(\Delta_5)\Gamma(\Delta_6)}{\Gamma(\Delta_5+\Delta_6)} = \Gamma(1+i\lambda)\Gamma(1-i\lambda) = \frac{\pi\lambda}{\sinh(\pi\lambda)} =: P(\lambda). \quad (2)$$

Proof. Equation (1) is Theorem 4.1 of [1], proved there from the celestial parameterization with all Jacobians computed, and verified to between 10^{-20} and 10^{-31} at five points with complex Δ . Equation (2) then follows from $\Gamma(2) = 1$ and the reflection formula, checked here to 10^{-41} . $\square$

The locus $\Delta_5 + \Delta_6 = 2$ is where the M dependence of (1) drops out, so it is the locus on which the cut is scale invariant. Nothing is localized there by a pole, and no residue is taken. This corrects the mechanism assumed throughout the earlier drafts.

2 Identities satisfied by the weight

Theorem 2.1 (Planck form, with cancelling zero-point energy). *With $n_B(y) = (e^y - 1)^{-1}$ and $E(x) = \frac{x}{2} \coth \frac{x}{2}$,*

$$P(\lambda) = 2\pi\lambda [n_B(\pi\lambda) - n_B(2\pi\lambda)] = 2\pi\lambda \sum_{n \geq 0} e^{-2\pi\lambda(n+\frac{1}{2})}, \quad 2E(\pi\lambda) - E(2\pi\lambda) = P(\lambda). \quad (3)$$

The zero-point terms cancel identically: $2 \cdot \frac{\pi\lambda}{2} - \frac{2\pi\lambda}{2} = 0$.

Proof. Both statements reduce to $\coth y - \coth 2y = 1/\sinh 2y$ after clearing factors. Verified at three values of λ to 3×10^{-39} or better. $\square$

Proposition 2.2 (Fourier pair).

$$P(\lambda) = \int_{\mathbb{R}} e^{i\lambda x} \frac{dx}{4 \cosh^2(x/2)}, \quad \frac{1}{2\pi} \int_{\mathbb{R}} P(\lambda) \cos(\lambda y) d\lambda = \frac{1}{4 \cosh^2(y/2)}. \quad (4)$$

Verified at two values to 1.7×10^{-41} .

Proposition 2.3 (Logarithmic expansion). *For $|\lambda| < 1$, $\log P(\lambda) = -\sum_{k \geq 1} \frac{(-1)^{k+1} \zeta(2k)}{k} \lambda^{2k}$, so the Taylor coefficients of $\log P$ are even zeta values.*

Remark 2.4 (Thermal reading, treated separately). Equation (3) admits a reading through the Bisognano–Wichmann theorem, in which P is the partition function of one oscillator at the modular temperature $\beta = 2\pi$ weighted by boost frequency. That structure, together with the Matsubara pole positions, the Weierstrass product, and the Mellin moment family $\mathbf{m}_s = \pi^{-(s+1)}(1 - 2^{-(s+1)})\Gamma(s+1)\zeta(s+1)$, is developed in [5]. The ladder moment $\mathcal{M}_1$ below is the $s = 1$ member of that family.

3 Ladder moments

Definition 3.1 (Ladder moments).

$$\mathcal{M}_L := \frac{1}{(2\pi)^L} \int_{\mathbb{R}_{>0}^L} \left[\prod_{j=1}^L P(\lambda_j) \right] \left[\prod_{j=1}^{L-1} P(|\lambda_j - \lambda_{j+1}|) \right] d\lambda_1 \cdots d\lambda_L, \quad (5)$$

with the empty product equal to 1 at $L = 1$.

Theorem 3.2 (First two moments). $\mathcal{M}_1 = \frac{1}{8}$ and $\mathcal{M}_2 = \frac{1}{90} = \zeta(4)/\pi^4$.

Proof. For $L = 1$, $\int_0^\infty \pi\lambda / \sinh(\pi\lambda) d\lambda = \pi/4$ by the classical value $\int_0^\infty x / \sinh x dx = \pi^2/4$, so $\mathcal{M}_1 = (\pi/4)/(2\pi) = 1/8$, exactly. For $L = 2$ the double integral was recomputed here by independent quadrature, giving 0.011111111111111111 against $1/90$ with relative error 1.8×10^{-26} . $\square$

Remark 3.3 (Status of these numbers). Definition 3.1 is a definition of convolution integrals of P . Theorem 3.2 is a fact about those integrals. Neither statement asserts that $\mathcal{M}_L$ normalizes an L -loop amplitude. The earlier draft made that assertion, and it is withdrawn. The bound $\mathcal{M}_L \leq (1/8)^L$, which follows from $0 < P \leq 1$ applied to the rung factors, remains true of (5) and says nothing about ultraviolet behaviour of amplitudes.

4 Retractions

The following claims from `haar_qg_paper_v2.1.5.1` are withdrawn. They are listed individually so that anyone holding those drafts can mark them.

1. “ **$P(\lambda)$ is the Plancherel measure of $\mathrm{SL}(2, \mathbb{C})$.**” False. The Plancherel density for the scalar principal series is proportional to λ^2 . The correct description of P is Theorem 1.1: a phase-space factor of the two-particle cut, equal to a ratio of Gamma functions on the shadow locus.
2. “**The flat Feynman spectral weight is replaced by $P(\lambda)$.**” Withdrawn. The companion derivation [1] reconstructs the ordinary loop integral from its cut by an unsubtracted dispersion relation, verified to relative error below 1.6×10^{-21} . No measure is replaced anywhere in that chain.
3. “ **$\mathcal{M}_L \leq (1/8)^L$, so the loop expansion is ultraviolet finite with no regularization and no counterterms.**” The inequality is true of Definition 3.1. The physical conclusion does not follow and is withdrawn. Divergences of the box appear exactly where standard methods put them; in [1] they are handled by an internal mass regulator, with the collinear logarithm confined to a subleading Mellin pole.
4. “ **$\mathcal{M}_1 = 1/8$ and $\mathcal{M}_2 = 1/90$ are the one- and two-loop measures.**” They are values of the integrals (5), and the rational values are a coincidence of the ladder graph topology at $L \leq 2$, as the earlier draft itself argued. The identification with loop normalizations is withdrawn.

5. **“The one-loop box is verified to fifteen significant figures with a 19.6-fold speedup.”** Withdrawn. The routine `box_shadow_1d` in `verify_qg_measure.py` integrates one Feynman parameter analytically and maps a second by a logistic change of variable, so `check_box` compares a Feynman-parameter integral against a reparameterization of itself. The comparison cannot test any celestial construction. That routine is deprecated and should not be cited.
6. **“The ultraviolet catastrophe of cavity radiation and the ultraviolet divergence of quantum gravity are the same disease with the same cure.”** Withdrawn as physics, and separately withdrawn in [5]. The Planck-form identity (3) and the thermal reading of Remark 2.4 stand as exact statements about the function P .
7. **Transcendentality claims tied to the digamma moment.** The identity $\frac{1}{2\pi} \int_{\mathbb{R}} P \operatorname{Re} \psi(\frac{1}{2} + \frac{i\lambda}{2}) d\lambda = \frac{1}{8} + \frac{1}{4} \psi(\frac{1}{2})$ is proved in [2]. The $\frac{1}{8}$ there arises from a full-line integral, while $\mathcal{M}_1 = \frac{1}{8}$ is a half-line integral, so reading the moment as “the one-loop measure plus a quarter of the digamma” overstates what has been shown.

5 What stands

Statement	Status	Check
$P = \Gamma(1 + i\lambda)\Gamma(1 - i\lambda)$ from phase space	proved	10^{-41}
Planck form, zero-point cancellation	proved	3×10^{-39}
Fourier pair with $\frac{1}{4} \operatorname{sech}^2(x/2)$	proved	1.7×10^{-41}
$\log P$ coefficients are even zeta values	proved	classical
Planck form and thermal reading	see [5]	
$\mathcal{M}_1 = 1/8$	proved	exact
$\mathcal{M}_2 = 1/90 = \zeta(4)/\pi^4$	verified	1.8×10^{-26}
P as Plancherel density	withdrawn	false
P as loop-measure replacement	withdrawn	false
UV finiteness without counterterms	withdrawn	false
Fifteen-digit box check	withdrawn	circular

References

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